

THEORY ASSIGNMENT - 01

Name:- Shweta Shinde

PRN:- 202501040359

TOPIC:- PYTHON BASICS: IF-ELSE STATEMENT

- What is if-else?

If the condition is true, one block of code runs.

If the condition is false, another block runs.

- What is If-Else?

Used for decision making

Works on conditions

SYNTAX

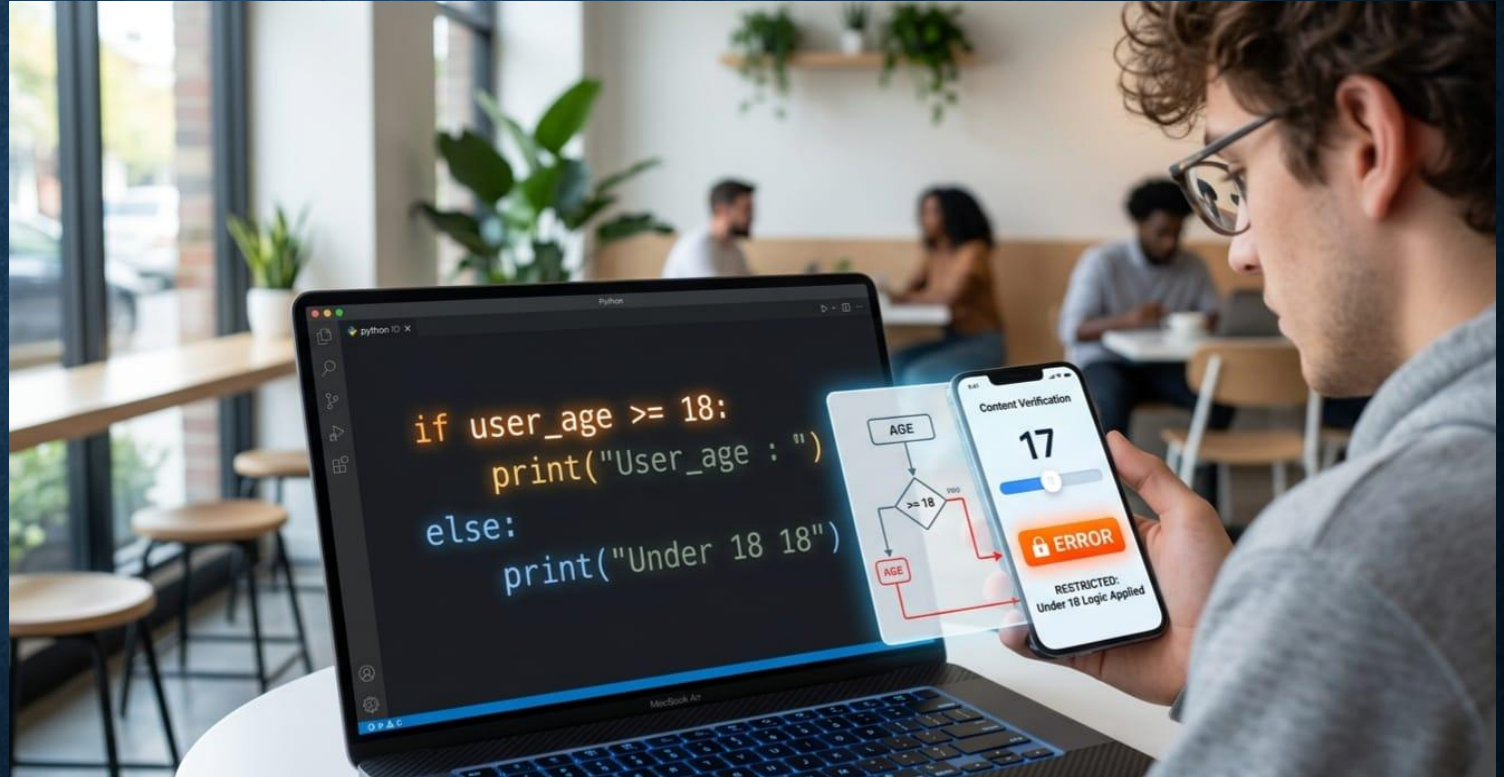
- Syntax

if condition:

statement

else:

statement



HOW IT WORKS

- How it Works

True → if block executes

False → else block executes

EXAMPLE

- Example

```
num = 5
```

```
if num > 0:
```

```
    print("Positive")
```

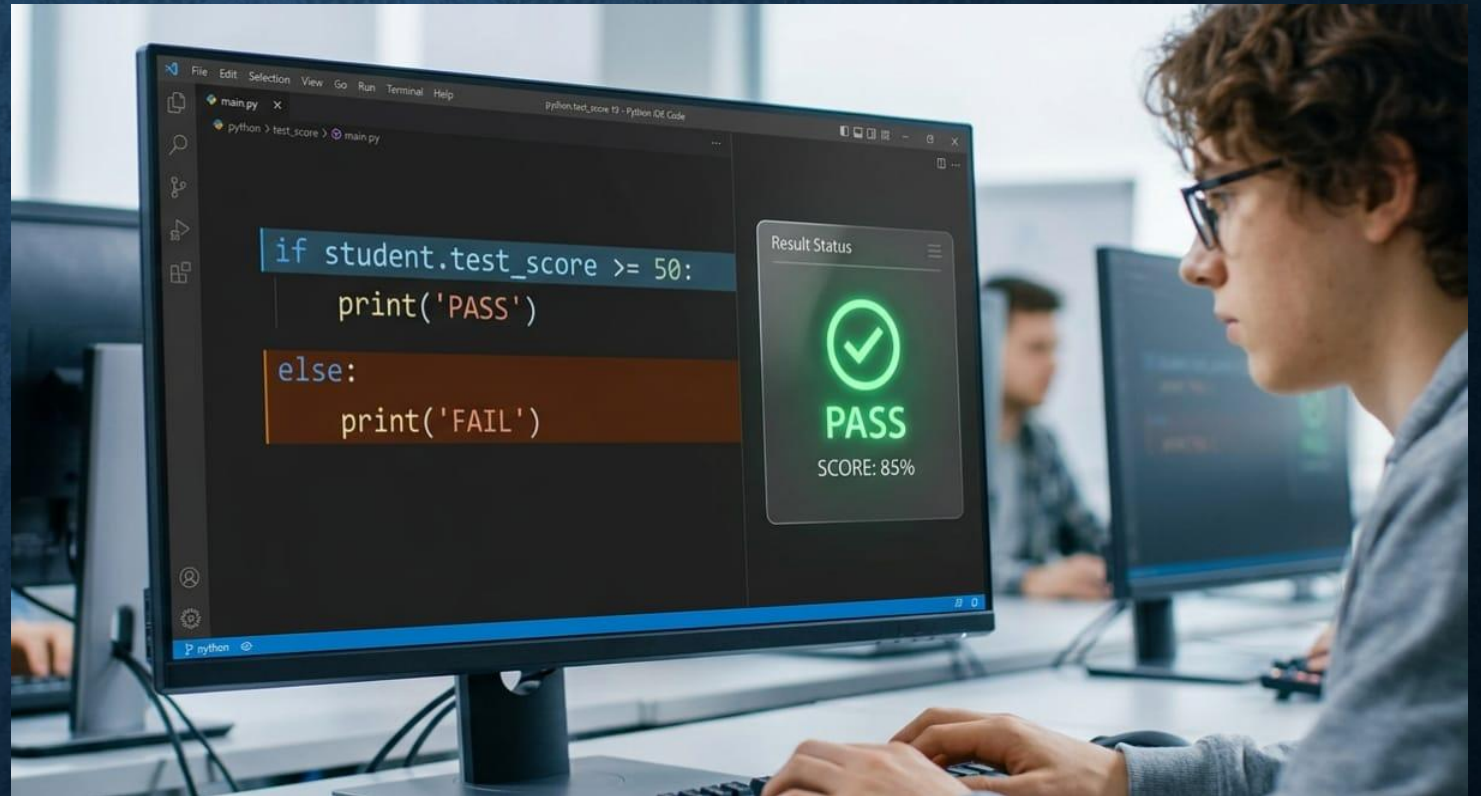
```
else:
```

```
    print("Negative")
```

- Output

Input: 5 → Positive

Input: -3 → Negative



REAL LIFE USE

- Login Systems
- Exam Result (Pass/Fail)
- ATM Machines
- Age Verification
- Online Shopping Discounts
- Traffic Signal System
- Hospital system

EXAMPLE :- ATM MACHINE

- ATM uses if-else for decision making
- It checks user balance before transaction
- Compares withdrawal amount with available balance

EXAMPLE

```
balance = 5000
```

```
withdraw = 2000
```

```
if withdraw <= balance:  
    print("Transaction Successful")  
else:  
    print("Insufficient Balance")
```



CONCLUSION

- Python provides simple syntax and is beginner-friendly
- Conditional statements like if-else allow programs to take decisions
- These concepts are used in real-world systems such as ATMs, apps, and websites
- Learning these basics helps in building strong programming skills

THANK YOU